

ET4382-SCPC Basic testbench 2025

Notes

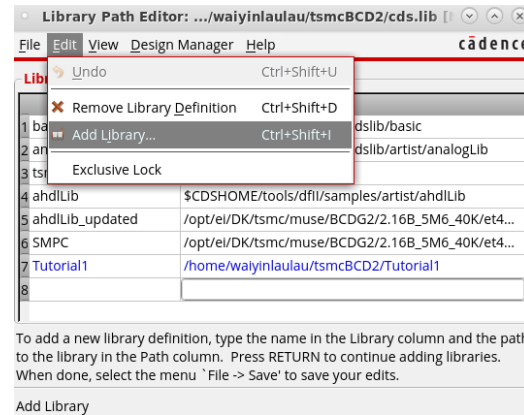
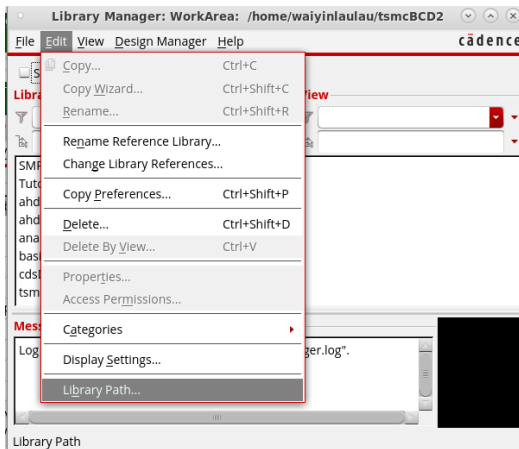
- This testbench is created according to SCPC HW1 (version: 24.03.2022) circuitA, as starting point for student to start SCPC HW.
- The circuit schematic and simulation expression are provided.
To run this simulation, student should read through the testbench and HW1 document, calculate necessary parameter and set up simulator.
- This testbench is only a reference for student to study, not exactly a grading guideline.

Steps set the testbench to your local(1)

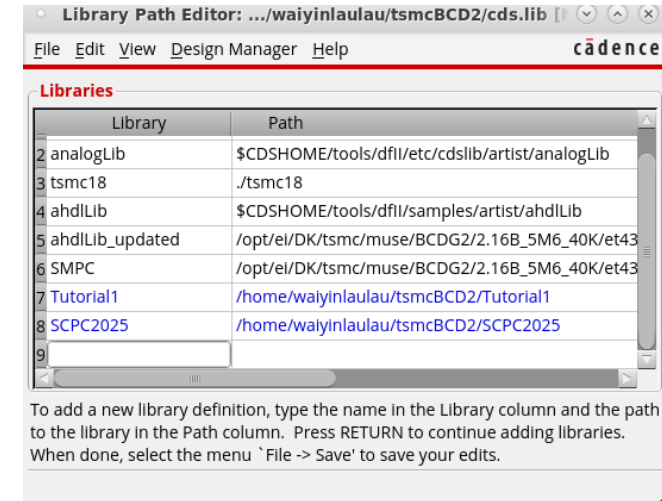
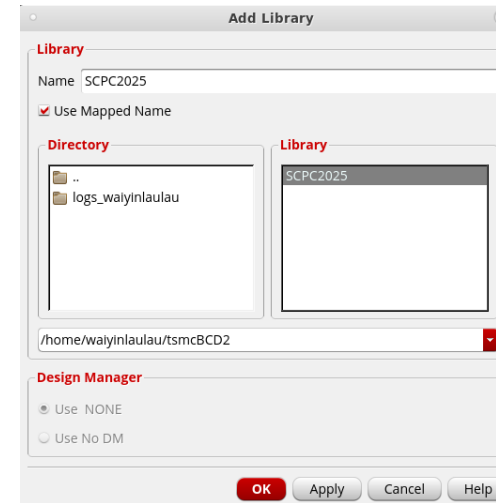
1. Copy the file from course server to your local with following command :
"cp -r /opt/ei/DK/tsmc/muse/BCDG2/2.16B_5M6_40K/et4382/SCPC2025
~/tsmcBCD"
 - "~/tsmcBCD" is the local folder which you created for your cadence files in tutorial1.
 - "SCPC2025" is the library created, and it contains the testbench.

Steps set the testbench to your local(2)

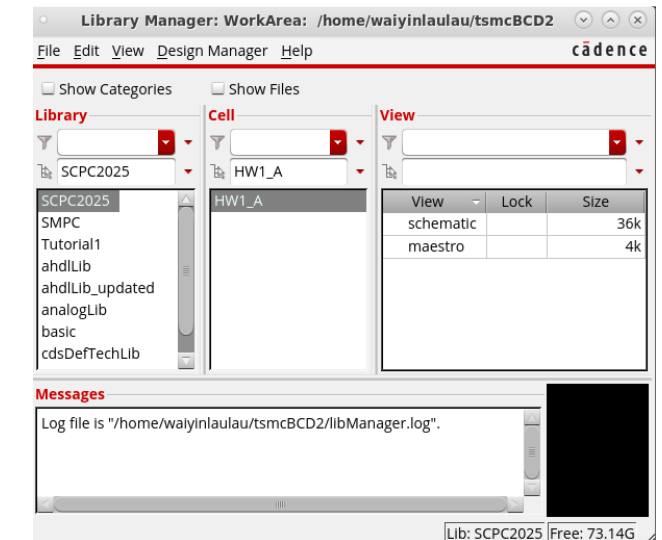
2. Add the library to your cadence



To add a new library definition, type the name in the Library column and the path to the library in the Path column. Press RETURN to continue adding libraries. When done, select the menu 'File -> Save' to save your edits.

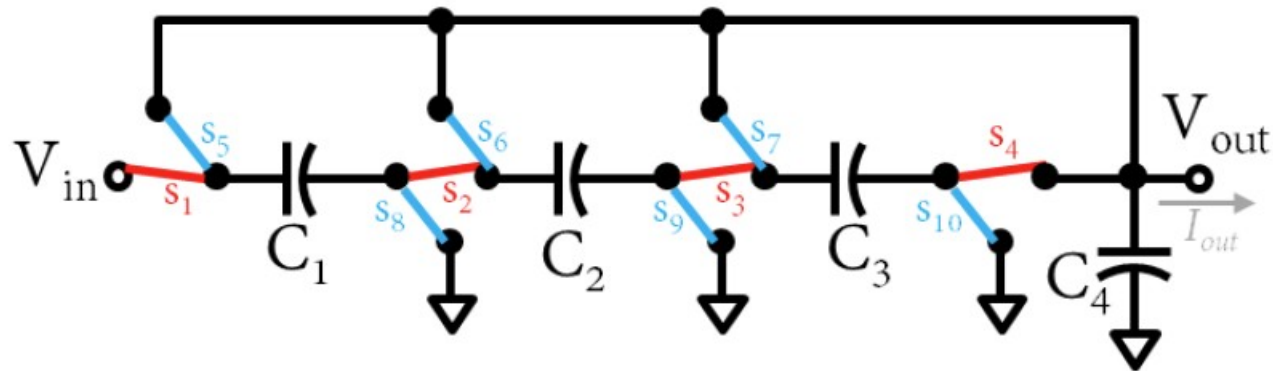


To add a new library definition, type the name in the Library column and the path to the library in the Path column. Press RETURN to continue adding libraries. When done, select the menu 'File -> Save' to save your edits.



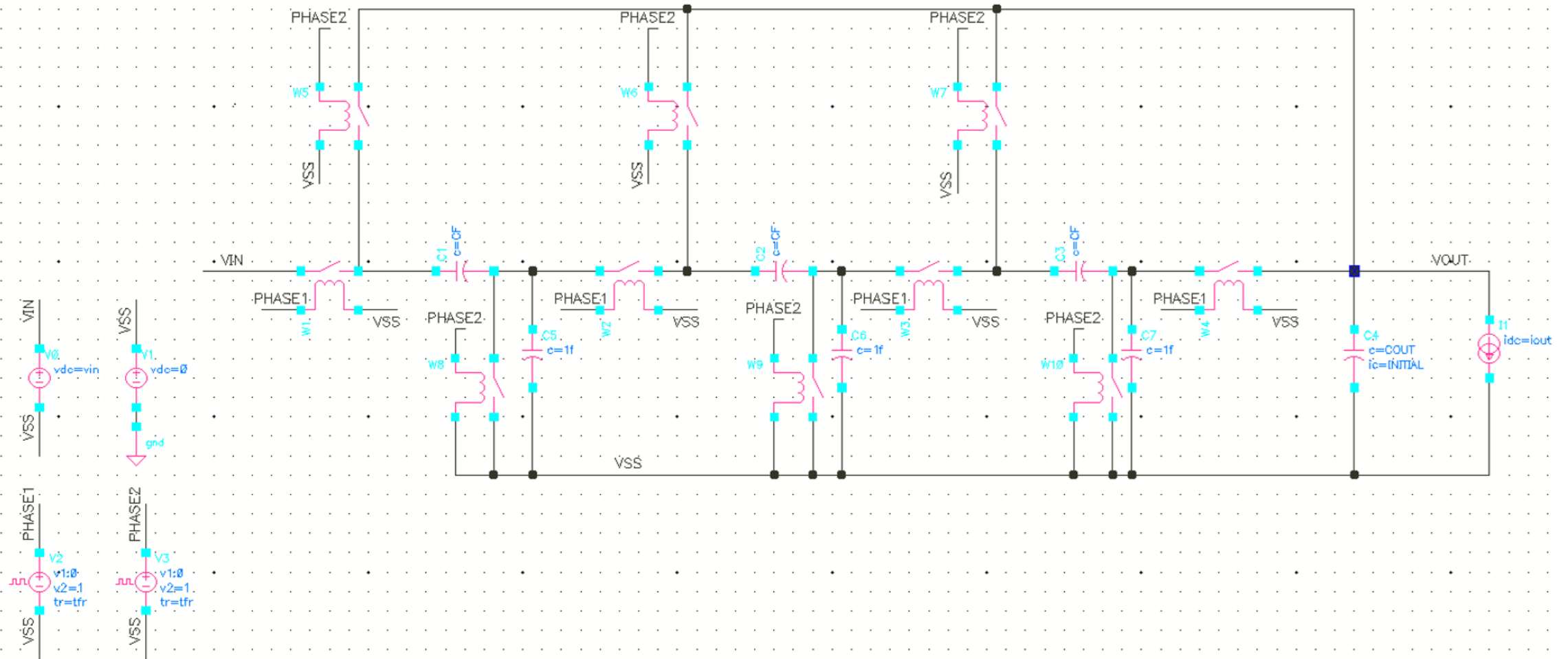
SCPC HW1:CircuitA(version: 24.03.2022)

Circuit A

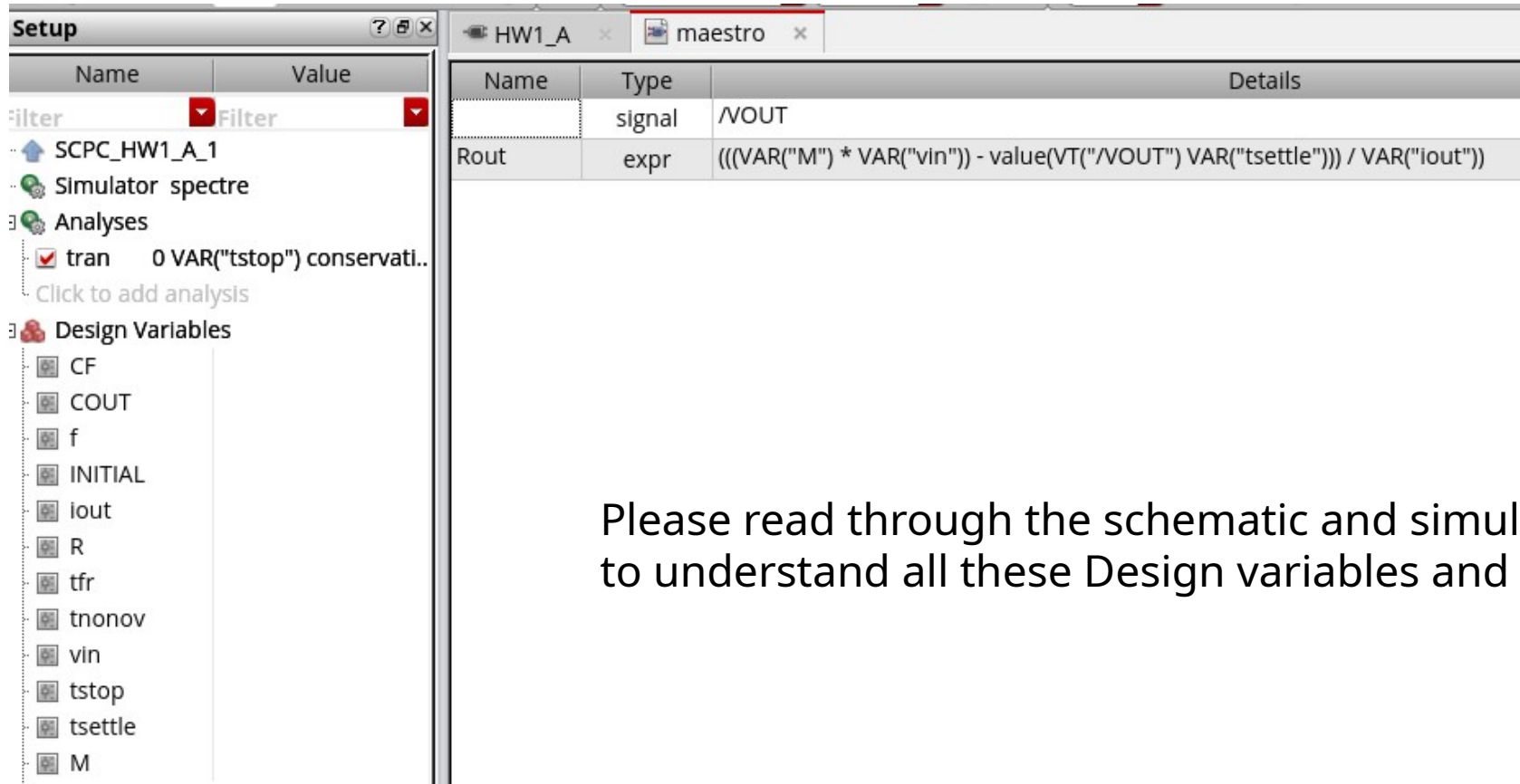


- switch is on in the 1st phase
- switch is on in the 2nd phase

Schematic



Simulation Setting



The screenshot shows the 'Setup' window of a simulation software. The window has two tabs: 'HW1_A' and 'maestro'. The 'maestro' tab is active. The window is divided into two main sections: 'Filter' and 'Design Variables'.

Filter Section:

Name	Value
filter	Filter

Design Variables Section:

Name	Type	Details
	signal	/VOUT
Rout	expr	$\frac{((\text{VAR}("M") * \text{VAR}("vin")) - \text{value}(\text{VT}("/VOUT") \text{VAR}("tsettle"))))}{\text{VAR}("iout")}$

Design Variables List:

- CF
- COUT
- f
- INITIAL
- iout
- R
- tfr
- tnonov
- vin
- tstop
- tsettle
- M

Please read through the schematic and simulation setting carefully to understand all these Design variables and fill them in